

Myoung Hoon Ha

Postdoctoral Researcher · Center for Neuroscience-inspired AI, KAIST

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RESEARCH INTERESTS

Predictive Coding and Iterative Inference, Temporal Sequence Modeling, Neural and Behavioral Latent-State Modeling, Neuroscience-inspired AI, Interpretable Learning

EDUCATION AND RESEARCH EXPERIENCE

- **Korea Advanced Institute of Science and Technology (KAIST)** Nov. 2019–Present
Postdoctoral Researcher, Center for Neuroscience-inspired AI / Advisor: Prof. Sang Wan Lee Daejeon, Korea
- **Seoul National University** Mar. 2011–Aug. 2019
Ph.D. in Electrical Engineering and Computer Science / Advisor: Prof. Byung-Ro Moon Seoul, Korea
 - Dissertation: *Evolution-based Online Meta Reinforcement Learning for Portfolio Optimization*
 - Concurrent with professional work at Optus Investment
- **Ajou University** Mar. 2002–Feb. 2010
B.S. in Information and Computer Science, Minor in Mathematics / Advisor: Prof. Tae-Sun Chung Suwon, Korea
 - *summa cum laude*; military service: 2004–2006

PUBLICATIONS

First-Author and Corresponding-Author Publications

- [1] **Myoung Hoon Ha**, Hyunjun Kim, Yoondo Sung, Youngha Jo, Min S. Kang, and Sang Wan Lee, *Stable and Scalable Deep Predictive Coding Networks with Meta Prediction Errors*, International Conference on Learning Representations (ICLR), 2026.
- [2] Hyunjun Kim and **Myoung Hoon Ha**^{*}, *Engram Memory Network: Brain-Inspired Prototype Explanations*, International Joint Conference on Neural Networks (IJCNN), 2026 (accepted) (* Corresponding author).
- [3] **Myoung Hoon Ha**, Seunggeun Chi, and Sang Wan Lee, *Learning to Escape: Multi-mode Policy Learning for the Traveling Salesman Problem*, IEEE International Conference on Evolving and Adaptive Intelligent Systems (IEEE-EAIS), 2024.
- [4] **Myoung Hoon Ha**[†], Hyung-gun Chi[†], Seunggeun Chi, Sang Wan Lee, Q. Huang, and K. Ramani, *InfoGCN: Representation Learning for Human Skeleton-based Action Recognition*, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 ([†] Equal contribution).
- [5] **Myoung Hoon Ha**, Seunggeun Chi, Sangyeop Lee, Yujin Cha, and Byung-Ro Moon, *Evolutionary Meta Reinforcement Learning for Portfolio Optimization*, Genetic and Evolutionary Computation Conference (GECCO), 2021.
- [6] **Myoung Hoon Ha** and Byung-Ro Moon, *The Evolution of Neural Network-based Chart Patterns: A Preliminary Study*, Genetic and Evolutionary Computation Conference (GECCO), 2017.
- [7] **Myoung Hoon Ha**, Sangyeop Lee, and Byung-Ro Moon, *A Genetic Algorithm for Rule-based Chart Pattern Search in Stock Market Prices*, Genetic and Evolutionary Computation Conference (GECCO), 2016.

Other Publications

- [8] Chamjin Joo, **Myoung Hoon Ha**, and Sang Wan Lee, *Rethinking Credit Assignment in Cooperative MARL via Interventional Reward Response*, Advances in Neural Information Processing Systems (NeurIPS), 2026 (accepted). OpenReview.
- [9] Jungwon Ryu, **Myoung Hoon Ha**, and Sang Wan Lee, *Generalizable Perceptual Embedding with Noise-Tuning Alignment*, Annual Computational Neuroscience Meeting (CNS), 2023.
- [10] Sangyeop Lee, **Myoung Hoon Ha**, and Byung-Ro Moon, *Understanding Features on Evolutionary Policy Optimizations: Feature Learning Difference between Gradient-based and Evolutionary Policy Optimizations*, ACM Symposium on Applied Computing (SAC), 2020.
- [11] Hansang Yun, **Myoung Hoon Ha**, and Robert Ian McKay, *VLR: A Memory-based Optimization Heuristic*, International Conference on Parallel Problem Solving from Nature (PPSN), 2014.

RESEARCH PROJECTS

•Stable and Scalable Predictive Coding Networks

Mar. 2023–Present

Principal Researcher

Center for Neuroscience-inspired AI, KAIST

- Led the core research on stability and scalability in deep predictive coding networks.
- Developed Meta-PCN, which diagnosed failure modes in deep predictive coding through dynamical mean-field analysis and introduced a principled stabilization framework.
- Demonstrated strong performance across CIFAR-10/100 and Tiny ImageNet while preserving local learning rules, leading to an ICLR 2026 publication.

•Neural Value Alignment (MSRA–IITP collaborative project)

Mar. 2023–Present

Principal Researcher

Center for Neuroscience-inspired AI, KAIST

- Designed and led a collaborative research proposal on neuroscience-grounded human–AI value alignment.
- Integrated predictive coding and neural-behavioral modeling into a unified research scope, milestone plan, and evaluation framework across project partners.

•Genie Brain: Brain-like Abstraction and Reasoning Engine

Oct. 2022–Oct. 2024

Project Lead & Principal Researcher

Center for Neuroscience-inspired AI, KAIST

- As sole contributor, developed the core algorithms for brain-inspired abstraction (System 1) and reasoning (System 2), from literature review and algorithm design to publication.
- Conducted collaborative research on language modeling, reformulating the problem as a curriculum-based self-supervised reinforcement learning task and contributing the resulting codebase to an industry partner.

•Neural and Behavioral Representation Learning

Nov. 2019–Oct. 2024

Research Scientist

Center for Neuroscience-inspired AI, KAIST

- Worked on joint modeling of neural and behavioral data, including rodent behavioral analysis and compact representation learning.
- Developed information-bottleneck-based objectives for extracting compact and generalizable latent representations from behavioral data.

•Brain-inspired Representation Learning and Interpretability

Nov. 2019–Oct. 2024

Project Lead / Research Scientist

Center for Neuroscience-inspired AI, KAIST

- Studied brain-inspired representation learning, including noise-tuning alignment, robustness-generalization trade-offs, and invariant representation learning for stochastic neural networks.
- Developed brain-inspired prototype-based explanation models, leading to *Engram Memory Network* (IJCNN 2026 accepted).

•Applied Reinforcement Learning and Sequential Decision Making

Mar. 2012–Mar. 2021

Technical Lead / Researcher

KAIST and Seoul National University

- Formulated and solved sequential decision-making problems in finance and optimization using reinforcement learning and evolutionary computation.
- Developed methods for portfolio optimization, real-time bidding, and combinatorial optimization, leading to publications in GECCO and IEEE-EAIS.

•Technical Guidance: Setting Questions for the SW Algorithm Test

Mar. 2013–Feb. 2021

Technical Advisor

Optimization Lab, Seoul National University

- Served as a technical advisor to an IT company, overseeing the design of software algorithm assessment problems.
- Verified problem correctness, supervised answer generation and timing validation, and supported assessment quality across multiple annual cycles.

•Development of ATM Cash Reserves Predictor

Mar. 2012–Feb. 2013

Graduate Researcher

Optimization Lab, Seoul National University

- Developed a data-driven system to forecast ATM cash withdrawal demand, supporting optimal cash reserve allocation and reducing operational costs.
- Conducted exploratory data analysis and built a user interface for the deployed software.

TEACHING

- Invited lecture series on reinforcement learning implementation (theory and hands-on coding), Hanyang University IDEC industrial training program (approx. 30 participants), Aug. 2022; Jul. 2023.
- Short lecture series on NLP and reinforcement learning (6 sessions), KAIST Center for Neuroscience-inspired AI internal seminar, Aug.–Sep. 2022.
- Guest lecture on applied reinforcement learning for the graduate course “Brain-inspired Artificial Intelligence,” KAIST, Fall 2021, 2022, 2023.

MENTORING

- Min S. Kang, KAIST, 2024–Present. Co-author on ICLR 2026.
- Hyunjun Kim, Seoul National University, 2024–2025. Co-author on ICLR 2026 and IJCNN 2026.
- Ye Mook Choi, KAIST, 2021–2023.
- Juno Kim, KAIST, 2020–2024.
- Seunggeun Chi, Seoul National University, 2019. Co-author on GECCO 2021 and IEEE-EAIS 2024.

INVITED TALKS

- *Disentangled Representation of Social Defeat Trace from Freely Moving Rodent Behavior*, Korean AI Association Annual Conference, 2021.
- *Evolutionary Policy Optimization for Control Problems in Computational Finance*, Center for Neuroscience-inspired AI Invited Seminar, KAIST, 2019.
- *A Deep Understanding of Artificial Neural Networks*, AI Research Council, Dept. of Mathematics, Ajou University, 2017.

SERVICE

- Reviewer: ICLR, IJCNN, GECCO, ACAIN.
- Session Chair, Microsoft Research Asia × KAIST BCS Joint Workshop, 2023.
- Session Chair, Neuroscience+AI, Korean AI Association, 2021, 2022.
- Session Chair, *Next-generation AI: Towards Human-level Intelligence*, 2020.

PRIOR INDUSTRY EXPERIENCE

- **Optus Investment** *May 2011–Nov. 2019*
Certified Investment Manager Seoul, Korea
 - Developed quantitative trading strategies using evolutionary algorithms and reinforcement learning, leading to publications in GECCO and related venues.
 - Built and operated algorithmic trading systems that translated research on pattern discovery and optimization into deployable financial software.
- **Pantech** *Jan. 2010–Dec. 2010*
Software Engineer Seoul, Korea
 - Developed Android home-screen applications and widgets for smartphones.

CERTIFICATION

- Certified Investment Manager, Korea Financial Investment Association (KOFIA), 2011.

SKILLS

Programming: Python, C/C++

ML/DL Frameworks: PyTorch, TensorFlow, JAX

Research Areas: Deep Learning, Reinforcement Learning, Evolutionary Computation, Computer Vision, Natural Language Processing, Representation Learning, Neuroscience-inspired AI

Tools: Git, L^AT_EX, Linux/Unix, SLURM (HPC)

Languages: Korean (native), English (professional)